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An LLM-Augmented Approach for Topic Modeling and Emotion Recognition in Vietnamese Online Education Platforms

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TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

ABBREVIATIONS

IT: Information Technology

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ABSTRACT

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CHAPTER 1

INTRODUCTION

1.1. Background

The past few years have witnessed a drastic advancement in technology, enhancing nearly every aspect of life. Education, in particular, is a field that has seen a significant shift toward online learning. For this reason, the need for a system that can understand student feedback and concerns has become increasingly important for the education system. Monitoring discussion forums and analyzing student feedback and comments is needed for gaining valuable insights, which can be used to tailor the teaching and management of the education system.

1. Challenges in Vietnamese Natural Language Processing (NLP)

Vietnamese is considered a low-resource language in natural language processing (NLP) research[1], making the implementation of language analysis systems such as sentiment analysis and topic modeling quite a difficult task. Historically, research on Vietnamese sentiment analysis (SA) has been hampered by the limited availability of suitable datasets[2]. Existing educational datasets for Vietnamese NLP often lack sufficient domain relevance and student slang, which presents a significant challenge for conducting research and improving the educational experience.

2. The demand for Advanced Text Analysis in Education

The two core methods of Text Analysis are Topic Modeling and Sentiment Analysis. These are essential for extracting meaning from the enormous volume of student-generated content, such as feedback, posts, and comments on online platforms.

Emotion Recognition (ER): is a higher-level approach than sentiment analysis[3]. In ER, text is not classified into general sentiments (such as positive, negative, or neutral) but into fine-grained emotions like joy, sadness, etc[3]. This detailed perspective will help gain a deeper understanding of student discussions. The foundational corpus for this work is the UIT-VSMEC (Vietnamese Social Media Emotion Corpus), which contains 6,927 human-annotated sentences across seven emotion labels[3].

Topic Modeling (TM) for context extraction: Topic modeling is a method for extracting "topics" from documents. However, in this research, this technique will serve a supporting role. The purpose is to extract and organize underlying themes from the student-generated content, which is necessary for interpreting and validating the detected emotions. By analyzing both the topics and the emotions, complex student feelings can be contextualized for a better understanding.

LLM Data Augmented Approach: The advancement of LLMs has brought up a new way to generate and labeling data. Many recent researches show that LLM has the capability to perform data diversity and generating by paraphrasing, response synthesis, counterfactuals, and many more[4]. This can be done by instructing LLM to generate synthetic examples exhibiting a given label by providing a labeled example and prompting the LLM to generate similar content[5].

1.2. Problem Statement

Implementing a reliable Fine-Grained Emotion Recognition for Vietnamese education context encounters several difficulties that can be solved by the LLM-Augmented approach:

1. **Poor performance and robustness in Fine-grained emotion recognition task:** BERT models achieve very high accuracy for basic sentiment polarity (e.g., positive, negative, neutral)[]. However, their performance drops noticeably when trying to classify documents into more detailed human emotions. For instance, the best classification model using CNN+word2vec on the UIT-VSMEC corpus only managed to achieve a weighted-F1 score of 59.74%[].
2. **Data scarcity and imbalanced dataset:** As mentioned before, Vietnamese is a low-resource language, hence domain-specific datasets are scarce[]. The dataset used in this research, UIT-VSMEC, suffers from a heavy imbalance in data distribution, as the "other" label occurs most frequently, and other emotions are lacking[]. This limits the model's robustness.

1.3. Scope and Objectives

Scope of research:

This study focuses on analyzing Vietnamese student-generated text from online platforms. The research specifically addresses two core NLP tasks:

1. **Fine-Grained Emotion Recognition**: The primary classification task involves identifying detailed emotions (e.g., enjoyment, sadness, fear, anger, disgust, surprise) within documents or text data.
2. **Topic Modeling**: The supporting task involves identifying and extracting a coherent thematic structure (topic) from the text to provide contextual support for the emotion analysis.

Objectives:

The objectives of this research are:

1. To leverage LLMs for controlled data augmentation to enrich and balance low-resource emotions.
2. To construct an LLM-BERT ensemble framework where the LLM acts as the selector to improve reliability and reduce misclassification. And as a result improve the accuracy of predictions.
3. To extract topics from text using LLMs to provide supporting context for the emotion extracted from text.
4. To evaluate the combined impact of augmentation and LLM-BERT ensemble on model performance across the dataset.

1.4. Assumption and Solution

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1.5. Structure of thesis

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CHAPTER 2

LITERATURE REVIEW/RELATED WORK

2.1. Review 1

Abc...

2.1.1. Sub review 1

Abc…

2.2. Review 2

2.2.1. Sub review 2

Abc...

…

CHAPTER 3

IMPLEMENTATION

3.1. User requirement analysis

Abc...

3.1.1. Sub 1

Abc...

3.2. System Design

Abc….

3.2.1. Database design

3.2.2. User Interface design

3.3. Abc

CHAPTER 4

EXPERIMENTS AND RESULTS

4.1. Experiments

Abc…

.

4.1.1. Abc

Abc….

Abc…

4.2. Results

Abc…

4.2.1. Result 1

Abc…

4.2.2. Result 2

Abc…

CHAPTER 5

DISCUSSION AND EVALUATION

5.1. Discussion

Abc…

5.2. Comparison

Abc…

5.3. Evaluation

Abc…

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1. Conclusion

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6.2. Future work

Abc…

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APPENDICES